MIL-E-17814E
21 March 1984
SUPERSEDING
MIL-E-17814D(YD)
19 July 1976

MILITARY SPECIFICATION

EXPANSION JOINTS, PIPE, SLIP-TYPE PACKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers guided packed slip-type expansion joints of metallic construction used to absorb dimensional movement caused by thermal expansion and contraction or other factors.

1.2 Classification. The expansion joints shall be of the following types, styles, classes, and forms as specified (see 6.2), and shall be of the size specified (see 6.2).

Type I - Ring-type packing.

Style 1 - Internally guided - externally guided.
Style 2 - Internally guided.

Class 1 - Single slip-tube.
Class 2 - Double slip-tube.

Type II - Ring-type packing with seal.

Style 1 - Internally guided - externally guided.
Style 2 - Internally guided.

Class 1 - Single slip-tube.
Class 2 - Double slip-tube.

Type III - Ring-type packing. Guides optional.

Class 1 - Single slip-tube.
Class 2 - Double slip-tube.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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Type IV - Internally-externally guided; injected semiplastic type packing.

Form A - Internal-external guides integral with joint body.
Form B - Internal-external guides of ring or bolt-on design.

Class 1 - Single slip-tube.
Class 2 - Double slip-tube.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-3 - Valve, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-F-3541 - Fittings, Lubrication.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B16.5 - Steel Pipe Flanges and Flanged Fittings.
B16.9 - Factory-Made Wrought Steel Buttwelding Fittings.
B31.1 - Power Piping.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A53 - Pipe, Steel, Black and Hot Dipped Zinc-Coated Welded and Seamless.
B650 - Electrodeposited Engineering Chromium Coatings on Ferrous Substrates.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish an expansion joint for first article inspection and approval (see 4.2.1 and 6.4).

3.2 Standard commercial product. The expansion joints shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the expansion joints being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3.1 Composition. Expansion joints shall consist of one or more of the following materials: Steel, ductile iron, or other alloys as specified (see 6.2 and 6.5).

3.4 Description. The expansion joint shall consist of a single or double slip-tube arranged to slide within the body of the joint. As specified

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herein, the expansion joint shall be provided with flanged or welded end connections, slip-tube guides, limit stops, tie-rods, stuffing boxes, packing glands, base, lubrication connections and fittings, service outlet, drain connection, vent connection, and the required bolts, studs, and nuts.

3.4.1 Type I.

3.4.1.1 Style 1. Joints shall have slip-tube guides located both internally and externally to the packing chamber. The packing shall be of the lubricated or self-lubricating ring type. A packing gland may be adjusted to seal the slip tube with service in operation. The service must be shut down to repack or add packing to this type joint. Limit stops shall be furnished in accordance with the manufacturer's standard practice.

3.4.1.2 Style 2. Joints shall have slip-tube guides located internally to the packing chamber. The packing shall be of the lubricated or self-lubricating ring type. A packing gland may be adjusted to seal the slip tube with service in operation. The service must be shut down to repack or add packing to this type joint. Limit stops shall be furnished in accordance with the manufacturer's standard practice.

3.4.2 Type II.

3.4.2.1 Style 1. Joints shall have slip-tube guides located both internally and externally to the packing chamber. Packing shall be of the lubricated or self-lubricating ring type. The joints shall be designed for the complete removal and replacement of packing while under full line pressure. Limit stops shall be furnished in accordance with the manufacturer's standard practice.

3.4.2.2 Style 2. Joints shall have slip-tube guides located internally to the packing chamber. Packing shall be of the lubricated or self-lubricating ring type. The joints shall be designed for the replacement of packing while under full line pressure. Limit stops shall be furnished in accordance with the manufacturer's standard practice.

3.4.3 Type III. Type III joints cannot have internal guides or limit stops which might interfere with the flow of viscous material. Guiding must be provided on the pipeline in accordance with the manufacturer's recommendations. Limit stops shall be furnished in accordance with the manufacturer's standard practice. The packing for this type of joint shall be of the lubricated or self-lubricating ring type. A packing gland may be adjusted to seal the slip tube with service in operation. The service must be shut down to repack or add packing to this type joint.

3.4.4 Type IV.

3.4.4.1 Form A. Joints shall have slip-tube guides of one piece formed or welded construction integral with the joint body. The primary packing for this type of joint shall be of the pressure injected, semiplastic type suitable for injection under full line pressure, although ring type packing may be used for sealing purposes of each end of the stuffing box. Limit stops should be furnished in accordance with the manufacturer's standard practice. This type joint must be removed from the line for repacking.

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3.4.4.2 Form B. Joints shall have slip-tube internal and external guides attached to the body or slip, held in a fixed position by a suitable recess or stop in the joint body. The primary packing for this type of joint shall be of the pressure injected, semiplastic type suitable for injection under full line pressure, although ring type packing may be used for sealing purposes of each end of the stuffing box. Limit stops should be furnished in accordance with the manufacturer's standard practice. The form B joint can be repacked without removing it from the line, but the service must be shut down before repacking.

3.4.5 Classes.

3.4.5.1 Class 1. Class 1 expansion joints shall have a slip-tube at only one end of the joint.

3.4.5.2 Class 2. Class 2 expansion joints shall have slip-tubes at both ends of the joint.

3.5 Performance. The joint shall be capable of absorbing the total axial expansion movement specified (see 4.5.2 and 6.2), and shall be designed to require the minimum of service maintenance when performing the function for which it is intended.

3.5.1 Service conditions. The expansion joint shall be suitable for operation under the following conditions of service, as specified (see 6.2):

a. Temperature-range to include installation, minimum, and maximum temperature.
b. Pressure-operating pressure and maximum shock pressure.
c. Type of material to be conveyed.
d. Velocity of material being conveyed.
e. Corrosive conditions.
f. Special conditions peculiar to installation.

3.5.2 Compression force. Unless otherwise specified (see 6.2), the force required to compress or extend the expansion joint shall not be greater than 1,000 pounds per inch of pipe diameter.

3.5.3 Hydrostatic pressure. Expansion joints up to and including 24 inches nominal pipe size shall be capable of withstanding, without damage, a hydrostatic pressure equal to 1-1/2 times the rated pressure of the joint. Joints larger than 24 inches shall be tested at a pressure to be specified by the procuring activity.

3.6 Construction. The expansion joints covered by this specification shall be designed and constructed in accordance with the engineering principles set forth in ANSI B31.1 where applicable.

3.6.1 Body. Unless otherwise specified (see 6.2), the body of the joint shall be made of ductile iron, cast steel, or fabricated steel, conforming to applicable ASTM standards for the pressure, temperature, and other service conditions. When the body is made of ductile iron, only the slip-tube end may be beveled for welding.

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3.6.2 Gland. Unless otherwise specified (see 6.2), the gland of the joint shall be made of ductile cast iron, cast steel, or fabricated steel, conforming to applicable ASTM standards for the pressure, temperature, and loading of this member.

3.6.3 Slip tube. The slip-tube shall be manufactured of steel complying with ASTM A53, type S, grade A or B. As a minimum (see 6.2), it shall be manufactured of Schedule 80 for all sizes to 14 inches inclusive and Schedule 60 for sizes 16 to 24 inches.

3.6.3.1 Chromium-plating. For types I, II, and III joints unless otherwise specified (see 6.2), steel sleeve or slip-tube surfaces shall be chromium plated by either decorative chromium per standard commercial practice or engineering chromium per ASTM B650. For type IV joints, the sliding slip-tube shall be plated with a minimum of 2 mils of engineering chromium per ASTM B650.

3.6.4 End connections. The end connections of the joint shall be provided with pipe flanges or welding nipples, as specified (see 6.2). Flanges shall conform to ANSI B16.5 and shall be of the size, material, pressure class, and facing specified. Welding nipples shall be beveled for welding to conform to ANSI B16.9 and shall be of the size, material, and nominal wall-thickness designation specified.

3.6.5 Limit stops. Limit stops may be of either the external or internal type, and shall be designed to withstand the full line pressure in all normal conditions of loading encountered during normal joint operation, including line pressure load resulting from failure of pipeline anchors. Tie rods may be used as external limit stops.

3.6.6 Guides. Slip-tube guides shall be designed to effectively maintain the slip-tube concentric with the stuffing box throughout the specified traverse. Type II joints, designed for repacking under full line pressure, shall have a piston-type internal guide provided with piston rings as a secondary pressure seal. When the joint body between the stuffing box and the rear of the slip-tube piston is properly vented through the by-pass connection, the piston rings shall be of sufficient number and suitable design to prevent the direct blow-back of line pressure through the stuffing box.

3.6.7 Stuffing box. The stuffing box shall be sufficiently deep and proportioned to hold ample packing and to assure a tight seal with minimum replacement or addition of packing. The design of the stuffing box shall be such that packing will not readily be extruded from the stuffing box and the lubricants used for sealing will not readily be extruded.

3.6.8 Packing glands. Packing glands shall be either the bolted type, or the threaded packing nut type, and shall be in accordance with the requirements for the type of fitting permitted by ANSI B31.1. The gland shall be designed to prevent seizing and scoring of the slip-tube. Packing pressure shall be readily and adequately adjustable and uniformly distributable. Joints of type IV shall be designed to prevent blow-back of the packing when an additional packing charge is introduced while the joint is under full line pressure.

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3.6.9 Base. When specified (see 6.2), class 1 expansion joints shall be provided with a base. The base for class 1 joints shall be designed primarily as a support for the joint and when specified, shall also be designed for use as a main anchor. All expected forces and moments for main anchor bases must be specified. Class 2 expansion joints shall be provided with a base suitable as an intermediate anchor. The base shall be of ducted iron, cast steel, or fabricated steel conforming to the applicable ASTM standards and shall be suitably attached to the joint, and drilled for the specified bolting.

3.6.10 Lubrication fittings. When packings require lubrication, fittings shall be provided. Lubrication fittings shall be in accordance with MIL-F-3541 or shall be the button-head type. When the joint is to be insulated, lubrication fittings shall be provided with extension nipples of suitable length for the thickness of the insulation specified (see 6.2).

3.6.11 Service outlet. When specified (see 6.2), a service outlet shall be achieved by the use of weldolet, threadolet, or saddleolets as specified. The service outlet shall permit unobstructed access, regardless of the position of slip-tubes.

3.6.12 Drain connections. When specified (see 6.2), each joint body shall be provided with a tapped and plugged drain connection. When type II joints are specified for repacking under full line pressure, the joint body shall be provided with a bypass connection, suitably tapped and plugged. The bypass connection shall be located between the rear of the slip-tube piston and the bottom of the stuffing box.

3.6.13 Bolting. Bolting required as an integral part of the joint shall be provided. Bolts, half studs, and accompanying nuts required as an integral part of the joint shall conform to the requirements of 3.6.

3.6.14 Packing. The packing shall be provided and installed by the manufacturer. Packing materials shall be suitable for the specified service and shall not react chemically with the components of the joint.

3.7 Drawings. When specified (see 6.2), a dimensional sketch or drawing in sufficient detail to describe the equipment the manufacturer proposes to furnish shall be submitted to the agency concerned with the bid proposal. These sketches or drawings shall be accompanied by a copy of the manufacturer's specifications and rating data in sufficient detail to signify compliance with the requirements of this specification.

3.8 Treatment and painting. Except for flanged faces and threaded connections all external surfaces subject to corrosion shall be treated and painted in accordance with the manufacturer's standard practice.

3.9 Repair or maintenance and accessories. Repair or maintenance parts and accessories as specified (see 6.2), shall be furnished.

3.10 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock washers required as an integral part of the joint shall be provided in accordance with good commercial practice and all bolts and nuts shall be completely fastened and tight.

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3.11 Workmanship. Workmanship shall be in accordance with the best commercial practice for this type of product. All expansion joints shall be clean and free of burrs, cracks, pits, porosity, and other defects affecting the serviceability of the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Inspection classification and requirements. Inspection shall be classified as follows:

a. First article inspection (see 4.3.1).
b. Quality conformance inspection (see 4.3.2).

4.2.1 First article inspection. First article inspection shall be performed on one expansion joint when a first article sample is required (see 3.1 and 6.2). This inspection shall include the examination of 4.4 and tests of 4.5. The first article may be a standard production item from the contractor's current inventory provided the unit meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. Any unit in a sample, as defined in 4.3.1, containing one or more defects, shall be rejected, and the lot represented by the sample shall be accepted or rejected as specified in MIL-STD-105.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All joints of the same type, style, and class offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

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4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 1.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-3 and an AQL of 1.5 percent defective.

4.4 Examination. Each joint selected shall be examined for compliance with the requirements specified in section 3 of this specification. The element of inspection shall encompass all visual and tactile examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects shall be cause for rejection.

4.5 Tests. Tests shall be conducted in accordance with 4.5.1 through 4.5.3. Any unit failing to pass these tests shall be rejected.

4.5.1 Hydrostatic test. Each test joint sample up to 24 inches in nominal pipe size shall be subjected to a hydrostatic pressure equal to 1-1/2 times the rated pressure of the joint. The pressure shall be maintained for 60 seconds. This test shall be performed with the joint extended. Joints larger than 24 inches nominal pipe size shall be subjected to pressure tests specified by the agency concerned (see 3.5.3).

4.5.2 Axial movement. Each test sample shall be tested to determine that the unit meets the specified axial-movement and axial-force requirement (see 3.5, 3.5.2, and 6.2).

4.5.3 Materials. Materials shall be tested in accordance with the applicable ASTM standard (see 3.6.3 and 6.5).

4.6 Packaging inspection. The preservation, packing, and marking shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. The preservation, packing, and marking shall be in accordance with MIL-V-3 with the level of cleaning, drying, preservation, and packaging and the level of packing as specified (see 6.2).

5.2 Preparation for shipment. Expansion joints shall be prepared for shipment in a manner that maintains the proper end-to-end dimensions required for installation of the joint. When no installation temperature is specified, the expansion joint shall be prepared for shipment with end-to-end dimensions in accordance with the manufacturer's standard practice. Each joint shall be accompanied by a complete set of instructions for compression or extension to the correct end-to-end dimension of the joint at the existing temperature during installation. These instructions shall be attached to the joint and shall be fully protected against moisture.

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6. NOTES

6.1 Intended use.

6.1.1 Service conditions. Slip-tube expansion joints are suitable for pressure-temperature service conditions in accordance with the ratings prescribed by the ANSI flange standards for the type of flanging materials used.

6.1.2 Types. No type of expansion joint listed in this specification is intended for operation without adequate provision for guiding and anchoring the system in accordance with the manufacturer's recommendations and in accordance with standard piping practices.

6.1.3 Sizes. Expansion joints most often are custom engineered and fabricated for specific applications. Because there is an infinite number of sizes that might be required for use in piping systems which include both standard and nonstandard pipe sizes, specific sizes are not cited in the classification paragraph of this specification (see 6.2).

6.1.3.1 Type I. The type I joint can have both internal and external slip-tube guides (style 1), or internal guides only (style 2). The type I joint ordinarily is supplied with standard ring-type packing. The service must be shut down in order to repack or add additional packing material. A packing gland may be adjusted to seal the slip tube with service in operation.

6.1.3.2 Type II. The type II joint can have both internal and external slip-tube guides (style 1), or internal guides only (style 2). This type of joint usually is supplied with standard ring type packing. This joint also must be installed with provision for venting so that it can be repacked while under full line pressure.

6.1.3.3 Type III. This joint cannot have internal guides or limit stops which might interfere with the flow of viscous material. Guiding must be provided on the pipeline in accordance with the manufacturer's recommendations. This type of joint shall be provided with limit stops to prevent over-extension of the slip in the event of anchor or guiding failure. A packing gland may be adjusted to seal the slip tube with service in operation. This type of joint is required for applications where the flowing media is either molten or highly viscous such as molten sulphur or asphalt or where large quantities of solid particles (such as cement) are in the flow stream. The service must be shut down to add packing or to repack this type joint.

6.1.3.4 Type IV. The type IV joint has slip-tube guides provided both on the internal (pressure) and external (atmospheric) side of the packing chamber. The external guide is either integral with or rigidly attached to the body. The packing for this joint is of the semiplastic type. Packing can be added while the joint is under full line pressure, but to completely repack the joint, the service must be shut down.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, style, form, class, and size required (see 1.2, 6.1.2, and 6.1.3).
c. First article if required for inspection and approval (see 3.1, 4.3.1, and 6.4).
d. Materials other than as specified (see 3.3.1 and 6.5).
e. Total axial expansion movement (see 3.5).
f. Service conditions (see 3.5.1).

(1) Temperature (degrees Fahrenheit): maximum and minimum operating temperature and superheat.
(2) Pressure (pounds per square inch gage), whether constant, pulsating, or shock.
(3) Fluid conveyed.
(4) Internal or external corrosive conditions if applicable.
(5) Size (nominal pipe size).

g. Compression force, if other than specified (see 3.5.2).
h. Whether the body, gland, and sleeve shall be constructed other than as specified (see 3.6.1, 3.6.2, and 3.6.3).
i. Whether joint shall be chromium plated (see 3.6.3.1).
j. Type end connection for flanged end connections specify ANSI pressure class, facing, and drilling required. For welding ends specify pipe schedule (or wall thickness) of nipples (see 3.6.4).
k. If base for support is required for class 1 expansion joints (see 3.6.9).
l. If base for main anchor is required for class 1 expansion joints and expected forces and moments (see 3.6.9).
m. Whether lubrication connections and fittings are required, and insulation thickness if joint is to be insulated (see 3.6.10).
n. Service outlet arrangement of outlet connection and type of connection if required (see 3.6.11).
o. If a drain connection is required (see 3.6.12).
p. If sketches or drawings are required (see 3.7).
q. Repair or maintenance parts, tools, and accessories required (see 3.9).
r. Test pressure for joints larger than 24-inch pipe size (see 4.5.2).
s. Level of cleaning, drying, preservation, and packaging and level of packing required (see 5.1).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

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6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete expansion joint. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Materials from which expansion joints are constructed may include, but are not limited to, the materials called out in the following:

ASTM A 36	Structural Steel.
ASTM A 53	Pipe, Steel, Black and Hot Dipped Zinc-Coated, Welded and Seamless.
ASTM A 106	Seamless Carbon Steel Pipe for High Temperature Service.
ASTM A 285	Pressure Vessel Plates, Carbon Steel, Low and Intermediate Tensile Strength.

6.6 Supersession data. Users are directed to Military Specification MIL-E-17813 on bellows expansion joints for the internally - externally guided permanent packless (bellows) seal joint, formerly referred to as type V in previous editions of this specification.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Air Force - 82

User activity:
Army - CE

Preparing activity:
Navy - YD

(Project 4730-0516)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

4730-0516

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-17814E	EXPANSION JOINTS, PIPE, SLIP-TYPE PACKED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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